

Project Report: Comprehensive Employee Management System

1. Project Overview

The Employee Management System is a robust, multi-role web application designed to centralize and streamline human resources, employee engagement, and system administration. The platform utilizes a role-based architecture, directing users to specific portals based on their designated roles: Employee, HR, or Admin.

The system incorporates a highly secure Authentication Flow, featuring a login screen equipped with a role selector, dual login methods (Password and OTP tabs), validation states, and a session timeout screen.

2. System Modules & Architecture

The application is divided into three primary portals, each tailored to specific operational needs:

A. Employee Portal (12 Modules)

The employee-facing interface is highly detailed, providing staff with self-service tools and real-time tracking:

- **Dashboard:** Features a welcome banner, four statistical cards, an attendance bar chart, quick actions, and a list of recent tasks.
- **Profile:** Organized into five editable tabs, including Personal, Employment (details and compensation), Skills (progress bars and certifications), Documents, and Emergency Contacts.
- **Attendance:** Contains an animated check-in/out card with a toggle, a monthly calendar grid, and a historical attendance table.
- **Leave Management:** Includes balance cards with progress bars, a usage bar chart, a history table, and an application modal featuring date pickers.
- **Payroll:** Renders detailed payslips showing earnings and deductions, net pay, YTD summaries, and an area trend chart.
- **Tasks:** Offers a filterable List view and a Kanban board with drag-column headers.
- **Performance:** Displays rating cards, score progress bars, goals tracking with completion states, and a review history table.
- **Chat:** Features a split layout for contacts and threads, a typing indicator animation, and real-time-style message bubbles.
- **Notifications:** Provides type-filtering tabs, unread indicators, and a left-border accent for unread messages.
- **Training:** Showcases course cards complete with progress bars and status badges.
- **Documents:** Provides a file list enabling document preview and download actions.
- **Settings:** Divided into five tabs covering Account, Security (2FA and sessions), Notifications (toggles), Appearance (theme toggle), and Language preferences.

B. HR/Manager Portal (6 Modules)

The HR portal equips managers with tools for oversight and recruitment:

- **Overview:** Displays dual charts, including an attendance bar chart and a department pie chart, alongside pending leaves and interview schedules.
- **Employee Management:** Features search and filter capabilities, full CRUD operations, and an employee profile modal.
- **Leave Approval:** Provides a dedicated workflow with quick approve or reject actions.
- **Recruitment:** Tracks job openings, manages the candidate pipeline, and includes an interview scheduler.
- **Departments:** Utilizes a card-based interface for department management.
- **Performance Reviews:** Incorporates a structured ratings matrix for employee evaluations.

C. Admin Dashboard (6 Modules)

The administrative portal ensures system health, deep analytics, and configuration:

- **Analytics:** Combines employee growth and payroll charts with system health monitoring.
- **User Management:** Displays role-based permissions utilizing a permissions matrix.
- **Payroll:** Offers views for salary structures and batch payroll processing.
- **Reports & Analytics:** Features five distinct tabs integrated with interactive Recharts.
- **System Settings:** Manages the permissions matrix, backup and restore functions, and overall company configuration.
- **Audit Logs:** Tracks user actions and provides severity filtering.

3. Technology Stack & Dependencies

Based on the project's internal directory structure and package dependencies, the system is built on a modern, performance-optimized JavaScript ecosystem:

- **Frontend Framework:** The application is built using **React** and **React DOM**.
- **Routing Architecture:** Client-side navigation is handled via **React Router**.
- **State Management:** Complex application state is managed using **React-Redux** alongside **Immer** for immutable state updates.
- **UI Components & Iconography:** User interface iconography relies heavily on **Lucide-React**.
- **Data Visualization:** Analytics and reporting modules render interactive charts using **Recharts**.
- **Styling & Processing:** CSS compilation and styling optimizations are processed through **PostCSS** and **LightningCSS**.
- **Core Utilities:** The project utilizes **es-toolkit** for math, string, and object manipulation, **Nanoid** for generating unique identifiers, and **EventEmitter3** for event handling.